Logics & Symbolic AI – cheat sheet

1 Propositional logic

- **Syntax** = manipulating symbols; **semantics** = meaning.
- Formula: symbol p, constants $T,\bot$; $(\neg F)$; $(A \bullet B)$.
- Syntactic/semantic: ¬/Not, ∧/And, ∨/Or, ⊃/⇒, ⊂/⇐, ≡/⇔, ↑/Nand, ↓/Nor.
- **Valuation** $v : F \to \{True, False\}$, consistent **(4):** **1)** $v(T) = True$ **2)** $v(\bot) = False$ **3)** $v(\neg F) = \text{Not } v(F)$ **4)** $v((A \bullet B)) = v(A) \blacksquare v(B)$.
- **Tautology**: $v(X) = True$ for every v.
- **Satisfiable** S: some v_0 makes all $X \in S$ True (SAT).
- X tautology $\iff \neg X$ not satisfiable.
- **Consequence** $S \models X$: every v satisfying S makes X True.
- **Key**: $S \models X \iff S \cup \{\neg X\}$ not satisfiable.
- Ex falso: $A, \neg A \in S \Rightarrow S \models X$. Monotony: $S \models X \Rightarrow S \cup \{A\} \models X$. Deduction: $S \cup \{X\} \models Y \iff S \models (X \supset Y)$.
- X tautology $\iff T \supset X$ tautology, since $v(T \supset X) = v(X)$.

2 CNF & resolution

- **Negation normal form**: ¬ only before symbols.
- $[X_1..X_n]$ = generalized disjunction ($v([]) = False$).
- $\langle X_1..X_n \rangle$ = generalized conjunction ($v(\langle\rangle) = True$).
- **CNF** $\langle C_1,..,C_n \rangle$, each **clause** $C_i = [\ldots]$.

	rewrite	negated		rewrite	negated
∧	$X \wedge Y$	$\neg X \vee \neg Y$	∨	$X \vee Y$	$\neg X \wedge \neg Y$
⊃	$\neg X \vee Y$	$X \wedge \neg Y$	⊂	$X \vee \neg Y$	$\neg X \wedge Y$
↑	$\neg X \vee \neg Y$	$X \wedge Y$	↓	$\neg X \wedge \neg Y$	$X \vee Y$
⊅	$X \wedge \neg Y$	$\neg X \vee Y$	⊄	$\neg X \wedge Y$	$X \vee \neg Y$

- Disjunction: $[..P..] \to [..A,B..]$.
- Conjunction: $[..P..] \to [..A..],[..B..]$.
- $\neg\neg P \to P$, $\neg T \to \bot$, $\neg\bot \to T$.
- **Resolution**: $[A,X,B] + [C,\neg X,D] \to [A,B,C,D]$.
- Proof of X: start $[\neg X]$, end with $[]$.
- For $S \vdash X$: start $[S_1],..,[S_n],[\neg X]$.
- X tautology $\iff X$ has a resolution proof.

Example $((A \supset B) \supset (A \supset C))$
$\langle [\neg(A \supset B), A \supset C] \rangle \to \langle [A \wedge \neg B, A \supset C] \rangle$
$\to \langle [A, \neg A, C], [\neg B, \neg A, C] \rangle \to \langle [\neg B, \neg A, C] \rangle$

- **Backward** chaining: from goal (resolution). **Forward**: from assumptions (Hilbert, modus ponens).
- $\models$ semantic (truth tables); $\vdash$ syntactic (proofs).
- **Sound**: $S \vdash F \Rightarrow S \models F$.
- **Complete**: $S \models F \Rightarrow S \vdash F$.

3 First-order logic

- **Terms**: variables, constants, functors $f(t_1..t_n)$. **Atomic formula**: predicate $p(t_1..t_n)$, arity n.
- Formulas: atomic, $\neg F$, $(F \circ G)$, $(\forall x)F$, $(\exists x)F$.
- **Functor or predicate?** Inside an argument ⇒ functor. At formula level (¬, ∧, ⊃, quantifier) ⇒ predicate.
- **Free** v **(4 rules):** **1)** occurs in atomic F **2)** in $(\diamond v')F$ iff free in F and $v \neq v'$ **3)** in $\neg F$ iff in F **4)** in $F_1 \circ F_2$ iff in F_1 or F_2.
- **Closed** formula: no free variable.

Semantics

- Domain $D \neq \emptyset$; interpretation I **(3):** **1)** $c^I \in D$ **2)** $f^I : D^n \to D$ **3)** p^I n-ary relation.
- Assignment A: $v^A \in D$. Model $M(D,I)$.
- **Attitude**: $p(t_1..)^{I,A} = True$ iff $(t_1^{I,A},..) \in p^I$.
- $(\forall x)F$: True for all B equal to A save for x; $\exists$: for one.
- **Valid**: true in every model (all A).
- F valid $\iff \{\neg F\}$ not satisfiable.
- Valid: $((\forall x)p(a,x)) \supset p(a,a)$. Not valid: $(\forall x)(\exists y)p(x,y)$.

Prenex (rename variables first)

- $\neg(\exists v)A \equiv (\forall v)\neg A$; $\neg(\forall v)A \equiv (\exists v)\neg A$.
- $((Qv)A \wedge B) \equiv (Qv)(A \wedge B)$, either side, $Q \in \{\forall, \exists\}$.
- $((\forall v)A \supset B) \equiv (\exists v)(A \supset B)$ (flips!).
- $((\exists v)A \supset B) \equiv (\forall v)(A \supset B)$ (flips!).
- $(A \supset (Qv)B) \equiv (Qv)(A \supset B)$ (kept).

Skolemization

- $..(\exists x_k)..F \to F(x_k / f(x_1..x_{k-1}))$, new functor.
- Arity = number of ∀ before it (0 = constant).
- Preserves satisfiability; put ∃ first if you can.
- $(\forall x)(m(x) \supset (\exists y)s(x,y)) \to (\forall x)(m(x) \supset s(x, f(x)))$.

Prove F valid by resolution (6 steps)

1. $\neg F$ in prenex form (rename variables).
2. Skolemize; **3)** drop ∀; **4)** CNF (optional).
5. Resolution + **unification** $(P(x), \neg P(c)$: $x = c)$.
6. $[] \Rightarrow \{\neg F\}$ unsatisfiable $\Rightarrow F$ valid.

- Propositional: sound, complete, **decidable**.
- FOL: sound, complete, **semi-decidable** (valid ⇒ proof found in finite time; not valid ⇒ may never stop).

4 Natural deduction (handout; ▷ = infer)

- Ax: $\triangleright \Gamma, p \vdash p$. $\bot$E: $\Gamma \vdash \bot \triangleright \Gamma \vdash p$.
- ¬I: $\Gamma, p \vdash \bot \triangleright \Gamma \vdash \neg p$. ¬E: $\Gamma \vdash p, \Gamma \vdash \neg p \triangleright \Gamma \vdash \bot$.
- Abs: $\Gamma, \neg p \vdash \bot \triangleright \Gamma \vdash p$ (proof by contradiction).
- →I: $\Gamma, p \vdash q \triangleright \Gamma \vdash p \to q$. →E: $\Gamma \vdash p \to q, \Gamma \vdash p \triangleright \Gamma \vdash q$.
- ∧I: $\Gamma \vdash p, \Gamma \vdash q \triangleright \Gamma \vdash p \wedge q$. ∧Eg / ∧Ed: $\Gamma \vdash p \wedge q \triangleright \Gamma \vdash p$ / $\Gamma \vdash q$.
- ∨Ig / ∨Id: $\Gamma \vdash p$ / $\Gamma \vdash q \triangleright \Gamma \vdash p \vee q$.
- ∨E: $\Gamma \vdash p \vee q, \Gamma, p \vdash r, \Gamma, q \vdash r \triangleright \Gamma \vdash r$.
- ∀I: $\Gamma \vdash F \triangleright \Gamma \vdash \forall x F$ (x not free in Γ). ∀E: $\Gamma \vdash \forall x F(x) \triangleright \Gamma \vdash F[t/x]$.
- ∃I: $\Gamma \vdash F(t) \triangleright \Gamma \vdash \exists x F(x)$. ∃E: $\Gamma \vdash \exists x F, \Gamma \vdash F \to G \triangleright \Gamma \vdash G$ (x not free in Γ, G).
- Handout names: g = left, d = right. Ex. $\vdash (A \to B) \to ((B \to C) \to (A \to C))$: from $\Gamma = \{A \to B, B \to C, A\}$, →E twice gives C, then →I ×3; term $\lambda f.\lambda g.\lambda x.g(fx)$.

5 Planning: STRIPS

- **State** describes the world, **actions** change it; planning = action sequence from initial to goal state. Predicates/Variables Capitalized, constants lowercase.
- `Move(X,Y)` Pre: `At(X), Level(low)` Post: `!At(X), At(Y)`
- STRIPS solver: backtracking, semi-decidable. **Sussman**: goals `On(a,b)`, `On(b,c)` clash.

6 Planning as SAT (Kautz & Selman 1992)

1. Add **time** as the last argument: `Move(a,b,1)`.
2. **Ground** every predicate (all argument values).
3. Grounded clause → boolean: `move_a_b_12`, `U-1-2`.
4. Write the **(5 kinds)** of constraints below.

- **1) Initial** ($t = 0$): every grounded fluent, + or ¬ (1 model).
- **2) Goal** ($t = N$): only goal facts (many models).
- **3) Problem**: e.g. table always clear, one action per step.
- **4) Action**: $a_t \Rightarrow (p_t \wedge c_{t+1})$.
- **5) Explanatory frame axioms (2 per fluent):**
  - $(\neg f_t \wedge f_{t+1}) \Rightarrow \bigvee$ actions at t that add f.
  - $(f_t \wedge \neg f_{t+1}) \Rightarrow \bigvee$ actions at t that delete f.
- SAT is decidable but N is fixed: increase N until satisfiable.

7 Formal Concept Analysis

- **Context** $K = (X, Y, I)$, $I \subseteq X \times Y$ (objects × attributes).
- $A^\uparrow = \{y \in Y \mid \forall x \in A : (x,y) \in I\}$ (shared attributes).
- $B^\downarrow = \{x \in X \mid \forall y \in B : (x,y) \in I\}$ (objects having B).
- $\emptyset^\uparrow = Y$, $\emptyset^\downarrow = X$.
- **Concept** (A,B) **(2, both!):** **1)** $A^\uparrow = B$ **2)** $B^\downarrow = A$. A extent, B intent.
- **Galois connection (4):** **1)** $A_1 \subseteq A_2 \Rightarrow A_2^\uparrow \subseteq A_1^\uparrow$ **2)** $B_1 \subseteq B_2 \Rightarrow B_2^\downarrow \subseteq B_1^\downarrow$ **3)** $A \subseteq A^{\uparrow\downarrow}$ **4)** $B \subseteq B^{\downarrow\uparrow}$.
- **Closure operator (3):** **1)** extensive $A \subseteq T(A)$ **2)** monotone **3)** idempotent $T(T(A)) = T(A)$.
- $(\cdot)^{\uparrow\downarrow}$ closes extents; $(\cdot)^{\downarrow\uparrow}$ closes intents.
- Always a concept: $(B^\downarrow, B^{\downarrow\uparrow})$ and $(A^{\uparrow\downarrow}, A^\uparrow)$.
- **Order**: $(A_1,B_1) \preceq (A_2,B_2) \iff A_1 \subseteq A_2 \iff B_2 \subseteq B_1$.
- **Meet** $\left(\bigcap X_t, (\bigcup Y_t)^{\uparrow\downarrow}\right)$; **Join** $\left((\bigcup X_t)^{\uparrow\downarrow}, \bigcap Y_t\right)$: infimum/supremum are concepts, so $(\mathscr{C}(K), \preceq)$ is a **complete lattice** (Hasse diagram).
- **Input**: context (objects = rows, attributes = columns). **Outputs (2):** **1)** concept lattice **2)** attribute implications (Duquenne–Guigues basis).
- Extensions: many-valued (X,Y,W,I); scaling, pattern structures; icebergs; triadic; relational.

Intersection method, row-based (Wille 1982; Stumme tutorial)
Rows = objects = *transactions*: intersecting the item sets of transactions gives the closed itemsets = intents. Theorem: every intent is an intersection of row intents $\{x\}^\uparrow$ (dually: columns). **(5 steps)**

1. Top to bottom: intersect each row intent $\{x\}^\uparrow$ with every row above it.
2. Result not yet a row ⇒ append it as a new row; continue until the last added row.
3. Add a full row (Y) if none. Now each row = one intent; its extent = intent$^\downarrow$.
4. Draw: fewest crosses on top; link each circle to the circles whose row it contains (immediate only).
5. Labels: attribute y at the highest circle containing it ($\{y\}^\downarrow$); object x at the lowest ($\{x\}^\uparrow$). Intent = labels above, extent = labels below.

Check the drawing (4): **1)** is it a lattice? **2)** one upper neighbour ⇒ has an attribute label **3)** one lower neighbour ⇒ has an object label **4)** g below $m \iff (g,m) \in I$.

- **NextClosure**: intents in lectic order $y_1 < .. < y_n$; try $y_i \notin B$ from last to first:
  - $C = ((B \cap \{y_1..y_{i-1}\}) \cup \{y_i\})^{\downarrow\uparrow}$; accept if C adds no attribute before y_i.
  - Start $\emptyset^{\downarrow\uparrow}$; stop at Y. $O(|X||Y|^2|C|)$.
- **Clarified**: no identical rows/columns (isomorphic lattice).
- **Reduced**: no object/attribute derivable from others.
- **Implication** $Z_1 \Rightarrow Z_2$ valid $\iff Z_1^\downarrow \subseteq Z_2^\downarrow$.
- Counterexample: object in $Z_1^\downarrow \setminus Z_2^\downarrow$.
- **Proofs**: $B \subseteq A^\uparrow \iff A \subseteq B^\downarrow$; $A^{\uparrow\downarrow\uparrow} = A^\uparrow$ ($\supseteq$ extensive, $\subseteq$ antitone); meet: $(B_1 \cup B_2)^\downarrow = B_1^\downarrow \cap B_2^\downarrow$.
- **Armstrong (3):** **1)** reflexivity **2)** augmentation **3)** transitivity.
- **Basis (2):** **1)** complete **2)** non-redundant. **Duquenne–Guigues**: pseudo-intents $P \Rightarrow P^{\downarrow\uparrow}$, minimal.
- **Slide example**: $x_1{:}y_1y_2y_3y_4$, $x_2{:}y_1y_3y_4$, $x_3,x_4{:}y_2y_3y_4$, $x_5{:}y_1 \Rightarrow 6$ concepts (intent / extent): $\emptyset$/all; $y_3y_4/x_1..x_4$; $y_2y_3y_4/x_1x_3x_4$; $y_1/x_1x_2x_5$; $y_1y_3y_4/x_1x_2$; Y/x_1. Clarify: $x_3 = x_4$, $y_3 = y_4$.

8 Association rules

- $\sigma(X)$ = number of transactions containing X.
- $s(X) = \sigma(X)/|T|$ (popularity); **frequent** if $s(X) \geq minSup$. Ex. $\sigma(\{newspaper, novel, film\}) = 2$, $s = 2/5$.
- $s(X \Rightarrow Y) = \sigma(X \cup Y)/|T|$.
- $c(X \Rightarrow Y) = \sigma(X \cup Y)/\sigma(X)$.
- Co-occurrence, not causality. Brute force: 2^d itemsets, $R = \sum_i C_d^i \sum_j C_{d-i}^j$ rules; cost = **passes**.
- **Independence**: $c(X \Rightarrow Y) = s(Y) \iff P(X \cup Y) = P(X)P(Y)$; $>$ positive correlation.
- **Lift** $= \frac{c}{s(Y)} = \frac{s(X \cup Y)}{s(X)s(Y)}$: $= 1$ independent, > 1 interesting. Tea⇒Coffee: $c = .75 < s(Coffee) = .8$.
- **A priori principle**: frequent ⇒ all subsets frequent; infrequent ⇒ supersets infrequent (support anti-monotone).
- **Apriori** (Agrawal & Srikant 1994) **(3):** **1)** pass 1: frequent items L_1 **2)** pass k: *join* L_{k-1} into candidates C_k, *prune* candidates with an infrequent subset, scan, keep $\geq minSup$ **3)** stop when none.
- Ex. $minSup = .4$: {film, classical, novel} pruned ({classical, novel} $s = \frac{1}{5}$); {film, newspaper, comics} kept ($s = \frac{2}{5}$).
- Passes: PCY 2 (hash), SON 2 (chunks), Toivonen 1–2 (sample), FP-Growth 2 (FP-tree, no candidates); FP-max: maximal sets.
- Rules from L: $A \Rightarrow L \setminus A$, $c = s(L)/s(A)$.

9 Fuzzy logic

- $\neg x = 1 - x$; Yager $\neg_w(x) = (1 - x^w)^{1/w}$.
- **T-norm (4):** **1)** commutative **2)** monotone **3)** associative **4)** neutral $T(a,1) = a$. So $T(a,0) = 0$.
- **(4 common)**: min (largest); product ab; Łukasiewicz $\max(0, a+b-1)$; drastic (smallest).
- Weak contradiction: $T(a, 1-a) \leq 0.5$.
- **T-conorm** $\bot(a,b) = 1 - T(1-a, 1-b)$ **(4)**: commutative, monotone, associative, identity $\bot(a,0) = a$.
- max (smallest); $1-(1-a)(1-b)$; $\min(a+b,1)$; drastic (largest).
- **S-implication** $(\neg x \vee y)$: $\max(1-x,y)$, $1-x(1-y)$, $\min(1-x+y,1)$.
- **R-implication** (residuum) $R_T(x,y) = \sup\{z : T(z,x) \leq y\}$:
  - $= 1$ if $x \leq y$;
  - else min: y; product: y/x; Łukasiewicz: $1-x+y$.
- $I(\neg a) = 1-a$, $I(a \wedge b) = T$, $I(a \vee b) = \bot$, $I(a \Rightarrow b) = R_T$.
- **Simple positive inference**: $P_1 \wedge .. \wedge P_n \Rightarrow_\phi C$.
  - Satisfied if $C \geq \phi(P_1..P_n)$.
  - Solution **(3):** **1)** $\geq$ initial values **2)** all rules satisfied **3)** no output can be smaller.
  - Iterate rules in any order until all satisfied.
  - With negation: may have no solution / oscillate.
- **Fuzzy set** $\mu : U \to [0,1]$ **(3 sets):** support $\mu > 0$; core $\mu = 1$; α-cut $\mu \geq \alpha$.
- α-capping $\min(\alpha, \mu(x))$.
- **Equality relation (3):** **1)** commutative **2)** reflexive $E(x,x) = 1$ **3)** transitive $T(E(x,y),E(y,z)) \leq E(x,z)$.
- Fuzzification $\mu(x) = \sup\{E(x,y) : y \in A\}$.
- **Defuzzification (3 methods):**
  - First of Maxima $\min\{c \mid \mu(c) \max\}$: safe, ignores other maxima.
  - Mean of Maxima: may land on a non-maximum.
  - Median: equal mass on both sides; may also miss.
- **Mamdani (4 steps):** **1)** fuzzify inputs **2)** $\alpha = T(\ldots)$ **3)** cap C at α **4)** defuzzify.

10 MAX SAT & PSL

- **Weighted MAX SAT**: world maximizing the total weight of satisfied formulas.
- Relaxation (Łukasiewicz, S-implication): clause degree $\min(\sum_i x_i, 1)$; negative literal gives $1 - y(s)$.
- **PSL**: max $\sum_c w(c) \min(\sum_{s+} y(s) + \sum_{s-}(1 - y(s)), 1)$, $y \in [0,1]$, $y(s) \leq v$.

Proof: Weighted MAX SAT is NP-hard (3 steps): **1)** take any SAT instance $C_1..C_m$ **2)** weight 1 each: satisfiable $\iff$ optimum $= m$ **3)** a polynomial algorithm would decide SAT (NP-complete) ⇒ P = NP. So: no polynomial algorithm unless P = NP (try all 2^n worlds); decision version is in NP ⇒ NP-complete.

Proof: PSL inference is polynomial (3 steps): **1)** $\ell_c(y) = \sum_{s+} y(s) + \sum_{s-}(1 - y(s))$ is affine **2)** $\min(\ell_c, 1)$ concave, positive-weighted sum concave, linear constraints ⇒ **convex** **3)** linear program: max $\sum_c w(c) z_c$ s.t. $z_c \leq \ell_c(y)$, $z_c \leq 1$, $0 \leq y \leq 1$ ⇒ polynomial (slides: hit and run).

11 Theorem proving & λ-calculus

- **Hilbert's program (3):** **1)** consistent **2)** complete **3)** decidable.
- **Gödel 1**: consistent system with arithmetic is incomplete.
- **Gödel 2**: it cannot prove its own consistency.
- Terms: variable x; abstraction $\lambda x.y$; application fx (left-associative).
- β: $(\lambda x.M)N \to M[N/x]$ (rename to avoid capture).
- **Strongly normalizing**: every reduction terminates.
- $(\lambda x.xx)(\lambda x.xx)$ never terminates; cannot be typed.

Church encoding (slides: number ↦ λ-term; below the λ-term of number m is written $\overline{m}$):

0	$\lambda f.\lambda x.x$	$n+1$	$\lambda n.\lambda f.\lambda x.f(nfx)$
1	$\lambda f.\lambda x.fx$	$m+n$	$\lambda m.\lambda n.\lambda f.\lambda x.mf(nfx)$
n	$\lambda f.\lambda x.f^n x$	mn	$\lambda m.\lambda n.\lambda f.\lambda x.m(nf)x$
True	$\lambda a.\lambda b.a$	¬	$\lambda p.p$ False True
False	$\lambda a.\lambda b.b$	∧;∨	$\lambda p.\lambda q.pqp$; $\lambda p.\lambda q.ppq$

- **Typed**: $\lambda x.y : \sigma \to \tau$; $fx : \tau$ needs $f : \sigma \to \tau$, $x : \sigma$.
- Simply typed λ is strongly normalizing (Curry).
- **Curry–Howard**: formula = type, proof = program.
  - →I = abstraction; →E (modus ponens) = application.
  - ∧ = product, ∨ = sum, ⊤ = unit, ⊥ = empty.
  - ∀ = dependent product, ∃ = dependent sum.
- Non-constructive proofs use excluded middle $P \vee \neg P$.

λ exercise sheet – solutions
Notation: m,n = λ-variables of the given term; $\overline{m}$ = the λ-term $\lambda f.\lambda x.f^m x$ of the *number* m, substituted for them.
Ex 1 *Give the reduced forms.*

- (1) $(\lambda x.x)y \to y$; (2) $(\lambda x.yx)z \to yz$.
- (3) $(\lambda x.\lambda y.x)(\lambda x.y)$: the free y would be captured ⇒ rename λy to λw: $\to \lambda w.\lambda x.y$.
- (4) $(\lambda x.xy)(\lambda z.z) \to (\lambda z.z)y \to y$.
- (5) $(\lambda x.\lambda y.xy(xy))(\lambda z.zz)$ $\to \lambda y.(\lambda z.zz)y((\lambda z.zz)y) \to \lambda y.yy(yy)$.

Ex 2 *What does $\lambda n.n(\lambda x.false)$true test about n?*

- Substitute $\overline{n}$: $\overline{n}(\lambda x.\text{false})\text{true} \to (\lambda x.\text{false})^n\text{true}$.
- $n = 0$: true. $n \geq 1$: last step $(\lambda x.\text{false})(\ldots) \to$ false. **Tests** $n = 0$.

Ex 3 *What is $\lambda m.\lambda n.nm$?*

- $(\lambda m.\lambda n.nm)\,\overline{m}\,\overline{n} \to \overline{n}\,\overline{m} = \overline{m}$ composed n times.
- $\overline{m}(\overline{m}f) = f^{m \cdot m} \Rightarrow \overline{n}\,\overline{m} = \lambda f.\lambda x.f^{m^n} x = \overline{m^n}$: **power** m^n. Check m=2, n=3: f^8.

Ex 4 *Why is $\lambda n.\lambda f.\lambda x.n(\lambda g.\lambda h.h(gf))(\lambda u.x)(\lambda u.u)$ the predecessor?*

- Substitute $\overline{n}$; let $S = \lambda g.\lambda h.h(gf)$: $\overline{n}\,S(\lambda u.x) = S^n(\lambda u.x)$.
- $S(\lambda u.x) = \lambda h.h((\lambda u.x)f) = \lambda h.hx$: first f *swallowed*.
- After $k \geq 1$ steps: $\lambda h.h(f^{k-1}x)$; apply to $\lambda u.u$: $f^{n-1}x$.
- So $\lambda f.\lambda x.f^{n-1}x = \overline{n-1}$ (and $n = 0$ gives $\overline{0}$).

12 Description logics (DL)

- **DL**: family of formal (logic-based) KR languages; world described by **concept descriptions** built from atomic concepts (classes), atomic roles (relationships) and individuals, with formal semantics.
- **Why DL, not FOL** (4): 1) variable-free syntax 2) easier to read 3) expressive yet decidable reasoning 4) most DLs = decidable fragments of FOL (FOL: expressive but undecidable).
- **DL system**: KB = **TBox** (terminology: concepts, roles) + **ABox** (assertions on named individuals); + description language, reasoning, applications, rules.
- **Inference services** (5): 1) TBox description satisfiable 2) subsumption (hierarchy) 3) ABox consistent (has a model) 4) individual instance of a concept 5) concept as query (retrieve individuals).

Syntax: A,B,C,D atomic concepts, R atomic role

- **AL** (6): $\top$ universal (Top) concept; $\bot$ bottom concept; $\neg A$ atomic negation (atomic only); $C \sqcap D$ intersection; $\forall R.C$ value restriction (all R-successors in C); $\exists R.\top$ limited existential quantification.
- Extensions (4): ALU union $C \sqcup D$; ALE full existential $\exists R.C$; ALC full negation $\neg C$; ALN number restrictions $\leq nR$, $\geq nR$.
- AL $\subseteq$ ALE $\subseteq$ ALC $\subseteq$ SHOIN $\subseteq$ SROIQ (OWL 2 DL).

Person $\sqcap$ Female	females who are persons
$\exists$hasChild.$\top$	has at least one child (any type)
$\exists$hasChild.Person	has a child who is a person
Person $\sqcap \forall$hasChild.Female	persons whose children are all female
Person $\sqcap \forall$hasChild.$\bot$	persons with no children
Person $\sqcap$ Female $\sqcap \geq 3$ hasChild	female persons with ≥ 3 children

Semantics: interpretation $\mathbb{I} = (\Delta^I, \cdot^I)$

- Domain Δ^I (non-empty set); interpretation function $\cdot^I$: atomic concept $A \mapsto A^I \subseteq \Delta^I$; atomic role $R \mapsto R^I \subseteq \Delta^I \times \Delta^I$.
- $(C \sqcap D)^I = C^I \cap D^I$; $(C \sqcup D)^I = C^I \cup D^I$; $(\neg C)^I = \Delta^I \setminus C^I$; $\top^I = \Delta^I$, $\bot^I = \emptyset$.
- $(\forall R.C)^I = \{x \in \Delta^I \mid \forall y, (x,y) \in R^I \rightarrow y \in C^I\}$.
- $(\exists R.C)^I = \{x \in \Delta^I \mid \exists y, (x,y) \in R^I \wedge y \in C^I\}$.
- $(\geq nR)^I = \{x \mid |\{y \mid (x,y) \in R^I\}| \geq n\}$ (same for $\leq$).
- Ex. Parent $\equiv$ Person $\sqcap \exists$hasChild.Person: $\forall x. Parent(x) \leftrightarrow (Person(x) \wedge \exists y.(hasChild(x,y) \wedge Person(y)))$.

DL $\leftrightarrow$ FOL (slide table)

$\top$, $\bot$	universal / empty	$\forall x.\top$, $\forall x.\bot$
$\neg C$	negation	$\neg C(x)$
$C \sqcap D$	conjunction	$C(x) \wedge D(x)$
$C \sqcup D$	disjunction	$C(x) \vee D(x)$
$C \sqsubseteq D$		$\forall x. C(x) \rightarrow D(x)$
$\exists R.C$	existential restr.	$\exists y. R(x,y) \wedge C(y)$
$\forall R.C$	universal restr.	$\forall y. R(x,y) \rightarrow C(y)$
$R_1 \circ R_2 \sqsubseteq R_3$	property chain	$\forall x,y,z. R_1(x,y) \wedge R_2(y,z) \rightarrow R_3(x,z)$

- Translation τ_x (Bloch): $\tau_x(\exists r.C) = \exists y, r(x,y) \wedge \tau_y(C)$; $\tau_x(\forall r.C) = \forall y, r(x,y) \rightarrow \tau_y(C)$; TBox $\bigwedge \forall x(\tau_x(C) \rightarrow \tau_x(D))$; ABox $C(a)$, $r(a,b)$.

TBox (terminology)

- Axioms (2): 1) inclusion $C \sqsubseteq D$ (C subclass of D) 2) equivalence $C \equiv D$ (same instances).
- I satisfies T iff I satisfies every axiom: I is a **model** of T, $I \models T$.
- Ex. $T = \{$Parent $\sqsubseteq$ Person, Mother $\equiv$ Woman $\sqcap \exists$hasChild.Person$\}$.
- Family TBox: Woman $\equiv$ Person $\sqcap$ Female; Man $\equiv$ Person $\sqcap \neg$Woman; Mother $\equiv$ Woman $\sqcap \exists$hasChild.Person; Parent $\equiv$ Father $\sqcup$ Mother; MotherWithoutDaughter $\equiv$ Mother $\sqcap \forall$hasChild.$\neg$Woman.

ABox (world description)

- Individuals a,b,c; **concept assertion** $C(a)$ (a belongs to C); **role assertion** $R(b,c)$ (c is a *filler* of R for b).
- Ex. MotherWithoutDaughter(Mary), Father(Peter), hasChild(Mary,Peter), hasChild(Peter,Harry), hasChild(Mary,Paul).
- Semantics (4): 1) I satisfies $C(a)$ if $a^I \in C^I$ 2) $R(a,b)$ if $(a^I, b^I) \in R^I$ 3) satisfies all assertions $\Rightarrow$ model of A 4) model of A w.r.t. T = model of T and of A.

Reasoning tasks for TBox (4)

- 1) **Satisfiability**: some model I of T with $C^I \neq \emptyset$.
- 2) **Subsumption** $C \sqsubseteq D$ or $T \models C \sqsubseteq D$: $C^I \subseteq D^I$ for every model I of T.
- 3) **Equivalence** $C \equiv_T D$ or $T \models C \equiv D$: $C^I = D^I$ for every model.
- 4) **Disjointness**: $C^I \cap D^I = \emptyset$ for every model.
- To subsumption (3): C unsat. $\iff C \sqsubseteq \bot$; equivalent $\iff$ both ways $\sqsubseteq$; disjoint $\iff C \sqcap D \sqsubseteq \bot$.
- To unsatisfiability (3): $C \sqsubseteq D \iff C \sqcap \neg D$ unsat.; equivalent $\iff C \sqcap \neg D$ and $\neg C \sqcap D$ unsat.; disjoint $\iff C \sqcap D$ unsat.

Reasoning tasks for ABox (4)

- 1) **Consistency**: A consistent w.r.t. T if $\exists I$ model of both; consistent = w.r.t. the empty TBox.
- 2) **Instance check**: $A \models C(a)$ if every I satisfying A satisfies $C(a)$; $A \models C(a) \iff A \cup \{\neg C(a)\}$ inconsistent.
- 3) **Retrieval**: given A, C: all a with $A \models C(a)$.
- 4) **Realization**: given a and a set of concepts: most specific C (minimal for $\sqsubseteq$) with $A \models C(a)$.

Extensions of ALC

- HappyMan $\equiv$ Human $\sqcap \neg$Female $\sqcap (\exists$married.Doctor$) \sqcap (\forall$hasChild.(Doctor $\sqcup$ Professor$)) \sqcap (\geq 2$hasChild $\sqcap \leq 4$hasChild).
- **Qualified number restriction** ≥ 2hasChild.Professor; **inverse roles** (hasChild / hasParent); **transitive roles** (hasAncestor); **subroles** (hasParent $\sqsubseteq$ hasAncestor); **concrete domains** (reals, integers, strings).

OWL (Web Ontology Language)

- Built on DL semantics: OWL 2 DL = SROIQ(D); reasoners HermiT, Pellet, FaCT++ (in Protégé); Python Owlready2, owlrl.
- Extension of RDF + RDF Schema. RDF/RDFS too limited (3): 1) cannot specify nature of relations (reflexivity…) 2) no reasoning 3) very limited logic.
- OWL adds: class connectors ($\sqcap, \sqcup, \neg$), cardinality, property characteristics, equivalence/equality of classes, properties, resources.
- Sub-languages (3): OWL Lite $\subset$ OWL DL $\subset$ OWL Full.
- Class declared (4 ways): 1) by name 2) by enumeration of individuals 3) by property restrictions 4) by $\sqcap/\sqcup/\neg$ of classes (anonymous class = individuals satisfying its definition). `owl:Thing` top, `owl:Nothing` subclass of all; `rdfs:subClassOf`; metaclass only in OWL Full; instances = the **extension**.

intersectionOf	$C_1 \sqcap \ldots \sqcap C_n$	$C_1(x) \wedge \ldots \wedge C_n(x)$
unionOf	$C_1 \sqcup \ldots \sqcup C_n$	$C_1(x) \vee \ldots \vee C_n(x)$
complementOf	$\neg C$	$\neg C(x)$
oneOf	$\{x_1\} \sqcup \ldots \sqcup \{x_n\}$	$x = x_1 \vee \ldots \vee x = x_n$
allValuesFrom	$\forall P.C$	$\forall y. P(x,y) \rightarrow C(y)$
someValuesFrom	$\exists P.C$	$\exists y. P(x,y) \wedge C(y)$
maxCardinality	$\leq nP$	$\exists^{\leq n} y. P(x,y)$
minCardinality	$\geq nP$	$\exists^{\geq n} y. P(x,y)$

- Restrictions (6): some ($\exists$), only ($\forall$), hasValue ($\ni$, "equals x"), exactly n, at most n, at least n.
- Property: domain, range, `rdfs:subPropertyOf` (inheritance). Characteristics (5):
  - **inverse** $P_1(X,Y) \iff P_2(Y,X)$ (isParentOf / isSonOf);
  - **transitive** $P(X,Y) \wedge P(Y,Z) \Rightarrow P(X,Z)$ (isTheAncestorOf);
  - **symmetric** $P(X,Y) \iff P(Y,X)$ (isBrotherOf);
  - **functional** $P(X,Y) \wedge P(X,Z) \Rightarrow Y = Z$ (hasAsMother);
  - **inverse functional** $P(Y,X) \wedge P(Z,X) \Rightarrow Y = Z$ (hasPasseportNumber).
- Protégé: **EquivalentTo** = necessary & sufficient; SubClassOf = necessary; disjoint classes; reasoner infers facts / detects inconsistencies (asserted vs inferred); SWRL rules; SPARQL queries.

Ontologies

- Gruber 1993: "explicit formal specifications of the terms in the domain and relations among them".
- Hudelot 2006: **formal, explicit specification of a shared conceptualization** (4): 1) conceptualization = view of a domain (concepts, definitions, relations) 2) explicit = types and constraints defined 3) formal = machine-understandable 4) shared = consensus of a group.
- Role: share, **reuse**, **interoperability**. Why build one (6): 1) precise, machine-processable definitions 2) separate domain from operational knowledge 3) common reference 4) shared understanding 5) reasoning/inference 6) explainable, trustworthy AI.
- Knowledge graph = triples (subject, predicate, object) = ABox level; ontology = schema/logic = TBox level.
- **Ontology structure** (quintuplet) $O := \{\mathscr{C}, \mathscr{R}, \mathscr{H}^{\mathscr{C}}, Rel, \mathscr{A}^O\}$:
  - $\mathscr{C}, \mathscr{R}$: disjoint sets of concepts and relations;
  - $\mathscr{H}^{\mathscr{C}} \subseteq \mathscr{C} \times \mathscr{C}$ taxonomy: $\mathscr{H}^{\mathscr{C}}(C_1, C_2) = C_1$ sub-concept of C_2;
  - $rel : \mathscr{R} \rightarrow \mathscr{C} \times \mathscr{C}$ (non-taxonomic relations), $dom(R) := \Pi_1(rel(R))$, $range(R) := \Pi_2(rel(R))$;
  - $\mathscr{A}^O$: axioms in a logical language (DL, FOL).
- **Knowledge base** (quadruplet) $KB := \{O, I, inst, instr\}$: O ontology, I individuals, $inst : \mathscr{C} \rightarrow 2^I$, $instr : \mathscr{R} \rightarrow 2^{I \times I}$; lexicon $L^{KB} := (L^I, J)$.
- Types (Guarino 98) (4): top-level (space, time, event), domain, task, application (roles of domain entities in an activity).
- Lightweight (E-R, UML) vs heavyweight (DL, FOL); standards RDF, RDFS, OWL.
- Life cycle (6): create/select, populate, validate (consistency), evolve, maintain, deploy.

Rule-based reasoning

- Modus ponens $\frac{A,\ A \rightarrow B}{B}$; modus tollens $\frac{\neg B,\ A \rightarrow B}{\neg A}$.
- Basic cycle (3): 1) detection (pattern matching / unification) 2) choice (first rule as Prolog, most specific, most useful; metarules) 3) execution (update KB).
- **Forward** chaining (data driven) / **backward** chaining (goal oriented: rules concluding the goal, prove their premises recursively) / mixed.
- Ex. R1 B,D,E→F; R2 D,G→A; R3 C,F→A; R4 C→D; R5 D→E; R6 A→H; R7 B→X; R8 X,C→A; B, C true, H? First rule: D,E,F,A,H. Last rule: X,A,H.

13 DL exam-style exercises

Prove $\forall r.(A \sqcap B) \sqsubseteq \forall r.A \sqcap \forall r.B$

- Take $x \in (\forall r.(A \sqcap B))^I$ and y with $(x,y) \in r^I$.
- Then $y \in A^I \cap B^I \subseteq A^I$, so $x \in (\forall r.A)^I$.
- Same for B, so x is in the intersection.
- FOL: $\forall x[\forall y(r \rightarrow A \wedge B) \rightarrow \forall y(r \rightarrow A) \wedge \forall y(r \rightarrow B)]$ valid. Converse holds too.
- $\exists r.(A \sqcap B) \sqsubseteq \exists r.A \sqcap \exists r.B$, *not* the converse.

Is Pedro a SuccessfulMan?

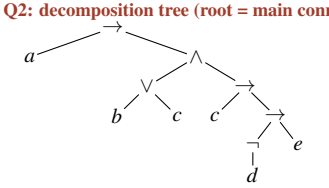
- SuccessfulMan $\equiv \neg$Female $\sqcap$ InBusiness $\sqcap \exists$married.Lawyer $\sqcap \exists$child.(StudiesAtUni $\sqcup$ Work).
- ABox: Pedro:$\neg$Female, Pedro:InBusiness, Mary:Lawyer, John:Works, married(Pedro,Mary), child(Pedro,John).
- Yes: Mary witnesses $\exists$married; John witnesses $\exists$child.
- $\equiv$ is sufficient; with only $\sqsubseteq$ it would not follow.
- Method: check each conjunct; witness for each $\exists$; all fillers for each $\forall$.

14 Example exam – full solutions

Propositional Q1: when is $F = a \rightarrow ((b \vee c) \wedge (c \rightarrow (\neg d \rightarrow e)))$ false?

- $X \rightarrow Y$ false iff $X = 1$, $Y = 0$: so $a = 1$ and $(b \vee c) \wedge H = 0$.
- Case 1: $b \vee c = 0$: $b = c = 0$, d, e free (4 rows).
- Case 2: $H = c \rightarrow (\neg d \rightarrow e) = 0$: $c = 1$, $\neg d = 1$, $e = 0$, so $d = e = 0$, b free (2 rows).
- *abcde*: 10000, 10001, 10010, 10011, 10100, 11100 (6).

Propositional Q2: decomposition tree (root = main connective)



Propositional Q3: CNF

- $\neg d \rightarrow e \equiv d \vee e$; $c \rightarrow (d \vee e) \equiv \neg c \vee d \vee e$.
- $F \equiv \neg a \vee ((b \vee c) \wedge (\neg c \vee d \vee e))$; distribute $\neg a$:
- $F \equiv (\neg a \vee b \vee c) \wedge (\neg a \vee \neg c \vee d \vee e) = \langle [\neg a, b, c], [\neg a, \neg c, d, e] \rangle$.

FOL Q1: function or predicate (and why)

- **Predicates** (operand of $\rightarrow, \neg, \vee, \wedge$, i.e. formula level): a/2, b/3, d/1, e/1.
- **Functions** (argument of another symbol, i.e. a term): c/2 in b, f/2 in e, g/1 in f.

FOL Q2: free variables

- $\forall x.\exists y.$ binds x, y only inside $(\neg b(\ldots) \vee d(x))$.
- x, y in $a(x,y)$ and in $e(f(x,g(y)))$ are outside that scope: free.
- z is never quantified: free. Answer: $\{x, y, z\}$.

Planning ($pX1_3$ = `posX-1-3`, $L2_3$ = `L-2-3`)

- Q1 initial state: $pX1_0 \wedge \neg pX2_0 \wedge \neg pX3_0 \wedge pY1_0 \wedge \neg pY2_0 \wedge \neg pY3_0$.
- Q2 goal (5 steps): $pX2_5 \wedge pY3_5$.
- Q3 frame axioms, posX at $t = 3$ (2 per value $\times$ 3 values = 6):
  - $(\neg pX1_3 \wedge pX1_4) \rightarrow L2_3$; $(pX1_3 \wedge \neg pX1_4) \rightarrow R1_3$
  - $(\neg pX2_3 \wedge pX2_4) \rightarrow R1_3 \vee L3_3$
  - $(pX2_3 \wedge \neg pX2_4) \rightarrow R2_3 \vee L2_3$
  - $(\neg pX3_3 \wedge pX3_4) \rightarrow R2_3$; $(pX3_3 \wedge \neg pX3_4) \rightarrow L3_3$
- Q4 Up at $t = 2$ (one per Y value = 3):
  - $U1_2 \rightarrow pY1_2 \wedge \neg pY1_3 \wedge pY2_3$
  - $U2_2 \rightarrow pY2_2 \wedge \neg pY2_3 \wedge pY3_3$
  - $\neg U3_2$ (precondition $Y < 3$ is false)

Fuzzy

- **Q1** $T(a,b) = \frac{a+b}{2}$ is **not** a T-norm: commutative, monotone hold; neutral fails $T(0.4,1) = 0.7$; associativity fails $T(T(0,0),1) = .5 \neq T(0,T(0,1)) = .25$.
- **Q2** $R_{\min}(0.3, 0.7) = 1$ since $0.3 \leq 0.7$; $\vee$ = conorm of min = max: $\max(1,0.1) = \mathbf{1}$.
- **Q3** Rules: $c \geq \min(a,b)$, $a \geq c$. Start $a = 0$ (no initial value), $b = .3$, $c = .1$.
  - Rule 1: $\min(0,.3) = 0 \leq .1$ OK. Rule 2: raise a to .1.
  - Rule 1: $\min(.1,.3) = .1 \leq .1$ OK. Stable: $\mathbf{a = .1, b = .3, c = .1}$.
- **Q4** First of Maxima, $d(\mu) = \min\{c \mid \mu(c)$ maximal$\}$:
  - $+$ always returns a point of maximal membership (safe); simple.
  - $-$ ignores the other maxima (best years 1960–2020 $\rightarrow$ only 1960); ignores the shape.
- **Q5** No (unless P = NP): reduce SAT (proof in section 10).
- **Q6** Yes: with $[0,1]$ values it is PSL, a convex problem (LP), polynomial.
- **Q7** Objective $f = a + 2\min(2 - a - b, 1) + 3b$:
  - if $a + b \leq 1$: $f = a + 2 + 3b$, best on $a + b = 1$: $f = 5 - 2a$;
  - if $a + b \geq 1$: $f = 4 - a + b$, maximal at $a = 0, b = 1$;
  - $\Rightarrow \mathbf{a = 0, b = 1}$, $f = 0 + 2 + 3 = 5$.

FCA: names $\rightarrow$ symbols; Q1

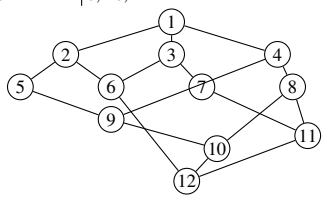
- $x_1..x_7$ = Frodo, Aragorn, Gandalf, Legolas, Saruman, Boromir, Gollum.
- $y_1..y_7$ = Middle-earth, Warrior, Magic, Immortal, Ring-carrier, Fellowship, Evil.
- **Q1 Yes**: rows $x_2 = x_6 \Rightarrow$ merge; no identical columns; y_1 full column (reducible).

FCA Q2: lattice by the intersection method

- New rows: $x_1 \cap x_2 = \{y_1 y_6\}$, $x_1 \cap x_5 = \{y_1 y_5\}$, $x_2 \cap x_5 = \{y_1\}$, $x_4 \cap x_5 = \{y_1 y_4\}$, $x_3 \cap x_5 = \{y_1 y_3 y_4\}$; add full row $Y \Rightarrow 12$ intents.

Table of concepts (every intent also contains y_1)

#	intent	extent	below
1	y_1	all 7	(top)
2	y_6	$x_1 x_2 x_3 x_4 x_6$	1
3	y_5	$x_1 x_5 x_7$	1
4	y_4	$x_3 x_4 x_5$	1
5	$y_2 y_6$	$x_2 x_3 x_4 x_6$	2
6	$y_5 y_6$	x_1	2, 3
7	$y_5 y_7$	$x_5 x_7$	3
8	$y_3 y_4$	$x_3 x_5$	4
9	$y_2 y_4 y_6$	$x_3 x_4$	4, 5
10	$y_2 y_3 y_4 y_6$	x_3	8, 9
11	$y_3 y_4 y_5 y_7$	x_5	7, 8
12	Y (all)	$\emptyset$	6, 10, 11



FCA Q3: closures ($^\downarrow$, $^\uparrow$, then $^\downarrow$ again gives the same extent)

- y_6: $\{y_6\}^\downarrow = \{x_1, x_2, x_3, x_4, x_6\}$; $^\uparrow = \{y_1, y_6\}$: concept 2.
- y_3: $\{y_3\}^\downarrow = \{x_3, x_5\}$; $^\uparrow = \{y_1, y_3, y_4\}$: concept 8.
- y_2: $\{y_2\}^\downarrow = \{x_2, x_3, x_4, x_6\}$; $^\uparrow = \{y_1, y_2, y_6\}$: concept 5.

FCA Q4: verify

- $(\{x_3, x_5\}, \{y_3, y_4\})$: **no**, $\{x_3, x_5\}^\uparrow = \{y_1, y_3, y_4\}$.
- $(\{x_1, x_2\}, \{y_6\})$: **no**, $\{y_6\}^\downarrow$ has 5 objects.

FCA Q5: implications ($|X| = 7$)

- $\{y_2, y_6\} \rightarrow \{y_4\}$: $\{y_2, y_6\}^\downarrow = \{x_2, x_3, x_4, x_6\}$, x_2, x_6 lack y_4: **invalid**; $s = \frac{2}{7}$, $c = \frac{2}{4}$.
- $\{y_5, y_6\} \rightarrow \{\neg y_7\}$: $\{y_5, y_6\}^\downarrow = \{x_1\}$, x_1 lacks y_7: **holds**; $s = \frac{1}{7}$, $c = 1$.

FCA Q6: association rules (AR) vs FCA

- Implication = AR with $c = 1$ ($Z_1^\downarrow \subseteq Z_2^\downarrow \iff \sigma(Z_1 \cup Z_2) = \sigma(Z_1)$); intents = closed itemsets, $s(B) = |B^\downarrow|/|X|$.
- FCA: small clean data, exact rules + hierarchy; AR: large noisy data, thresholds. E.g. $\{y_2\} \Rightarrow \{y_6\}$ $c = 1$ (implication); $\{y_6\} \Rightarrow \{y_2\}$ $c = \frac{4}{5}$ (AR only).